AI POWERED STOCK MARKET INSIGHT GENERATOR

INDIVIDUAL THESIS

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Individual component: Explainable AI-Based Financial Market Risk Impact Engine

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DECLARATION AND SUPERVISOR CERTIFICATION

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Abstract

Retail investors in the Colombo Stock Exchange (CSE) need to interpret stock-specific behaviour together with wider market conditions, yet a machine-learning risk label alone may be difficult for a non-expert user to understand. This thesis presents an Explainable AI-Based Financial Market Risk Impact Engine, developed as an individual component of the wider AI Powered Stock Market Insight Generator for the CSE. The component classifies the current risk condition of a selected stock as LOW, MEDIUM, or HIGH using nine inputs: Close Price, Volume, MA10, MA50, Volatility, Gold Price, Oil Price, VIX, and an encoded stock identifier. A persisted Random Forest classifier with 200 decision trees performs the classification. SHAP TreeExplainer is used to identify the factors associated with an individual prediction, while a separate Gemini 2.5 Flash layer converts the structured risk output and top factors into plain-language text without determining the risk class. The prototype integrates Python Flask, MongoDB, Selenium, yfinance, and a React/Vite dashboard. The archived evaluation artefact shows complete separation of the three prototype classes on one reported split, and the saved model assigns approximately 88.18% of its impurity-based feature importance to Volatility. These results are interpreted cautiously because the target labels were created from volatility cut-offs while Volatility is also an input feature. The study therefore contributes an end-to-end explainable risk-support workflow and, at the same time, documents the methodological changes required before the prototype can be treated as a robust future-risk forecasting system.

Keywords: Financial Market Risk, Explainable AI, Random Forest, SHAP, Colombo Stock Exchange

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LIST OF ABBREVIATIONS

Abbreviation	Meaning
AI	Artificial Intelligence
API	Application Programming Interface
CORS	Cross-Origin Resource Sharing
CSE	Colombo Stock Exchange
CSV	Comma-Separated Values
HTTP	Hypertext Transfer Protocol
JSON	JavaScript Object Notation
LLM	Large Language Model
MA	Moving Average
ML	Machine Learning
REST	Representational State Transfer
RF	Random Forest
SHAP	SHapley Additive exPlanations
UI	User Interface
VIX	CBOE Volatility Index
XAI	Explainable Artificial Intelligence

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1 INTRODUCTION

1.1 Overall Project Context

The wider research project, AI Powered Stock Market Insight Generator for the Colombo Stock Exchange, is intended to bring several forms of stock-market analysis into one investor-facing platform. The project context is motivated by a practical problem: retail investors can access increasing amounts of market information, yet the information is often distributed across separate charts, indicators, news sources, and analytical tools. The broader solution therefore aims to organise analytical outputs so that they can be inspected together rather than as isolated technical results.

Within the group solution, different components address different aspects of market intelligence. The present thesis is limited to the financial-market risk component. It does not attempt to replace the full group platform or to provide personalised investment advice. Instead, it produces a structured risk result that can be consumed by the wider dashboard together with other project outputs.

1.2 Individual Component Context

The individual component is titled Explainable AI-Based Financial Market Risk Impact Engine. It combines stock-specific indicators with selected global market indicators, classifies the current risk condition of a selected stock into LOW, MEDIUM, or HIGH, explains the model result using SHAP, and presents a plain-language explanation through an investor-facing web interface. The component currently supports Hemas Holdings PLC (HHL), John Keells Holdings PLC (JKH), and Chevron Lubricants Lanka PLC (LLUB) in the dashboard.

The component intentionally separates three responsibilities. Random Forest determines the risk class, SHAP identifies influential features for an individual result, and Gemini converts the structured output into a simpler explanation. This separation is central to the design because the natural-language model is not allowed to silently replace the machine-learning decision.

1.3 Problem Statement

A retail investor may observe a change in stock price or trading volume without being able to determine whether the current market condition should be treated as relatively low, moderate, or high risk. A machine-learning label alone does not fully solve this problem because the user may still be unable to understand why the model produced that label. The research problem is therefore to build an end-to-end component that combines stock-specific and global indicators, classifies the current risk state, identifies the main factors associated with an individual result, and communicates the result in a form that can be understood by a non-expert user.

1.4 Research Gap

The first gap is interpretability. Many stock tools provide prices, trends, or predictions, while the reasoning behind a model output remains unclear to the user. The second gap is contextual integration: stock-level features can be viewed separately from wider market indicators, but the user benefits from seeing both in one workflow. A third gap is local explanation. Global feature importance indicates which variables matter to a model overall, but it does not explain one selected prediction. Finally, a practical integration gap remains in bringing CSE data, historical feature calculation, machine learning, local XAI, and plain-language presentation into one component.

1.5 Aim and Objectives

Aim: To design, implement, and evaluate an Explainable AI-Based Financial Market Risk Impact Engine that classifies the current risk condition of selected CSE stocks as LOW, MEDIUM, or HIGH using stock-specific and global indicators, while explaining the main factors associated with each result.

- To collect and prepare historical and current stock-market data required for risk analysis.
- To engineer Close Price, Volume, MA10, MA50, and Volatility as stock-specific features.
- To integrate Gold Price, Oil Price, and VIX as selected global market indicators.
- To implement a Random Forest classifier for LOW, MEDIUM, and HIGH risk-state classification.

- To apply SHAP to identify influential features for individual predictions.
- To generate a plain-language explanation from the structured risk and SHAP outputs without allowing the language model to determine the risk class.
- To expose the analytical workflow through a Flask REST API and integrate it with the React/Vite dashboard.
- To evaluate classification behaviour, explainability, implementation quality, integration, and methodological limitations.

Table 1.1: Research Objectives

No.	Objective
1	To collect and prepare historical and current stock-market data required for risk analysis.
2	To engineer Close Price, Volume, MA10, MA50, and Volatility as stock-specific features.
3	To integrate Gold Price, Oil Price, and VIX as selected global market indicators.
4	To implement a Random Forest classifier for LOW, MEDIUM, and HIGH risk-state classification.
5	To apply SHAP to identify influential features for individual predictions.
6	To generate a plain-language explanation from the structured risk and SHAP outputs without allowing the language model to determine the risk class.
7	To expose the analytical workflow through a Flask REST API and integrate it with the React/Vite dashboard.
8	To evaluate classification behaviour, explainability, implementation quality, integration, and methodological limitations.

1.6 Research Questions or Hypotheses

- How can stock-specific indicators and selected global market indicators be combined to classify the current risk condition of selected CSE stocks?
- How effectively does the persisted Random Forest classifier separate the prototype LOW, MEDIUM, and HIGH classes in the available evaluation artefacts?

- Which features contribute most strongly to the model globally and to selected individual classifications according to SHAP?
- How can technical model explanations be converted into plain-language investor-facing explanations without changing the underlying risk decision?
- What methodological and implementation limitations must be addressed before the prototype can be treated as a robust real-world financial decision-support system?

1.7 Scope Limitations and Assumptions

The dashboard scope is limited to HHL, JKH, and LLUB. The model-development dataset is wider and contains 33 symbols from Sri Lanka, the United States, the United Kingdom, and India. The study assumes that the persisted model and encoder are the versions used by the prototype. Gold, Oil, and VIX are used as contextual indicators rather than as direct causal explanations of a CSE stock. The current risk classes are operational prototype labels built from volatility cut-offs and are not universal market standards. The system is therefore described as a current risk-state classifier, not as a validated predictor of future investment losses.

1.8 Individual Contribution

The individual contribution is the design and implementation of the financial-risk workflow: feature preparation, Random Forest inference, class mapping, SHAP-based local explanation, global-indicator integration, plain-language explanation, Flask API delivery, and the risk section of the React dashboard. The contribution is primarily one of integration, explainability, and user-facing interpretation rather than invention of a new machine-learning algorithm.

1.9 Thesis Structure

Chapter 2 reviews the literature and related techniques. Chapter 3 explains the research methodology and evaluation approach. Chapter 4 describes the product and solution design. Chapter 5 documents implementation, integration, and testing. Chapter 6 presents the available results and validation evidence. Chapter 7 discusses the meaning and limitations of the findings. Chapter 8 considers commercialisation and sustainability. Chapter 9 concludes the study and outlines future directions.

2 LITERATURE REVIEW AND RELATED WORK

2.1 Review Method and Scope

The literature review was scoped around four themes directly related to the component: financial-market risk and volatility, machine-learning classification, explainable artificial intelligence, and investor-facing decision support. Sources were selected to provide theoretical grounding for the implemented techniques rather than to claim that the exact prototype thresholds are established market standards. The review also distinguishes between methods used for prediction and methods used only for explanation or presentation.

2.2 Domain and Theoretical Foundations

Financial-market risk is closely associated with uncertainty and variation in returns. Engle [8] provided a major statistical foundation for time-varying variance in financial series. This thesis does not implement an ARCH-family forecasting model; instead, volatility is used as an engineered feature and as the basis of the current operational class labels. Moving averages provide short- and longer-term references for price behaviour, while volume represents trading activity.

VIX is widely interpreted as a measure of expected market volatility and uncertainty [7]. Gold and crude oil are included as broader market-context variables. In this thesis they are not treated as direct causes of a CSE stock's risk class. Their purpose is to provide the classifier and the explanation layer with a small amount of global context alongside the stock-specific indicators.

2.3 Related Research

Random Forest, introduced by Breiman [3], combines multiple decision trees and is suitable for nonlinear classification problems. Its tree structure is also compatible with efficient SHAP explanations. Interpretability literature emphasizes that predictive performance alone does not answer whether a model decision can be understood by a user [4], [5]. Lundberg and Lee [6] introduced SHAP as a unified approach for assigning feature contributions to individual predictions. These ideas motivate the model-and-explanation separation used in this component.

Investor-behaviour research also supports the need for careful presentation. Barber and Odean [1] discuss patterns in individual investor behaviour, while Bouteska and Regaieg [2] examine the relationship between investor sentiment and market volatility. Such work does not directly define this component, but it supports the broader requirement that investor-facing tools should present uncertainty and explanation responsibly.

Table 2.1: Literature Themes and Relevance

Source / Theme	**Main relevance**	**Gap relative to this component**
[1] Barber and Odean	Investor behaviour and decision patterns	Does not provide an ML risk engine.
[2] Bouteska and Regaieg	Investor sentiment and volatility	Not a CSE-specific explainable classification workflow.
[3] Breiman	Random Forest ensemble learning	Classification method, not a complete financial XAI product.
[4], [5] Interpretability literature	Model interpretation principles	Motivates explanation beyond accuracy.
[6] Lundberg and Lee	SHAP feature attribution	Provides local feature-level explanation.
[7] Whaley	VIX interpretation	Supports VIX as a market-uncertainty indicator.
Proposed component	CSE-facing risk classification + local XAI + plain language	Integrates classification, SHAP, global context, and dashboard delivery.

2.4 Existing Systems Products and Techniques

Existing market tools commonly provide charts, moving averages, price alerts, or summary indicators. Machine-learning research may provide classification or forecasting results, but the research output is often separated from a usable explanation interface. At the technique level, a single Decision Tree is easy to inspect but can be unstable; Random Forest improves robustness by aggregating many trees. Global feature importance is useful for model-level interpretation, while SHAP is better suited to explaining a selected prediction. A natural-language layer can improve readability, but it must be prevented from changing the model result or fabricating confidence.

2.5 Gap Synthesis and Conceptual Framework

Table 2.2: Research Gap Synthesis

Gap	Limitation	Response in this component
Risk-focused output	Charts and trends may not provide a direct risk state.	Explicit LOW / MEDIUM / HIGH prototype risk output.
Combined context	Single-source interpretation can be narrow.	Stock indicators + Gold + Oil + VIX.
Local explainability	Global importance does not explain one observation.	SHAP local attribution for individual predictions.
Plain-language interpretation	Technical XAI may be difficult for novice users.	Gemini rewrites structured results without making the decision.
CSE-oriented integration	Analytical components are often isolated.	CSE data, historical store, ML API, XAI, and React dashboard connected end-to-end.

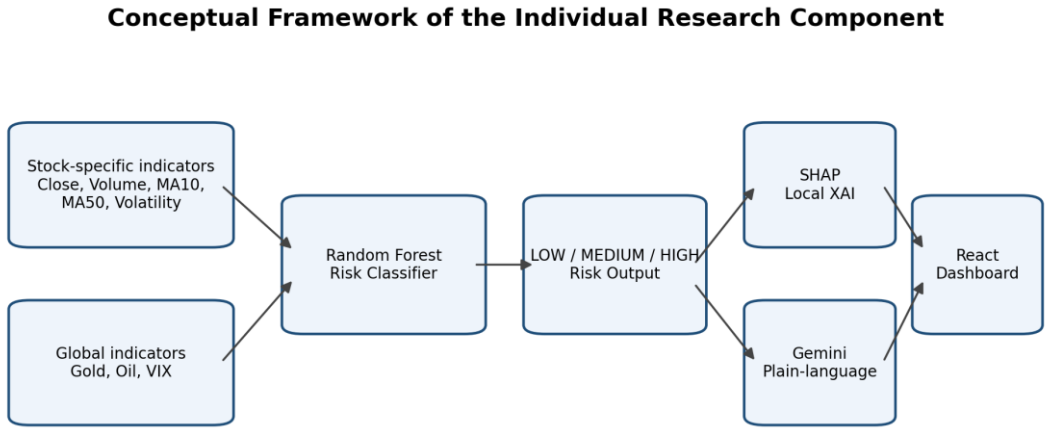


Figure 2.1: Conceptual Framework of the Individual Research Component

3 RESEARCH METHODOLOGY

3.1 Research Design

The study follows an artefact-oriented research design that combines machine-learning experimentation with software implementation. The research artefact is the explainable risk engine. The work proceeds from requirement identification and data preparation to feature engineering, target construction, model inference, explanation, integration, and validation. The purpose is not only to report a classifier but also to evaluate how the classifier is embedded into a transparent end-to-end workflow.

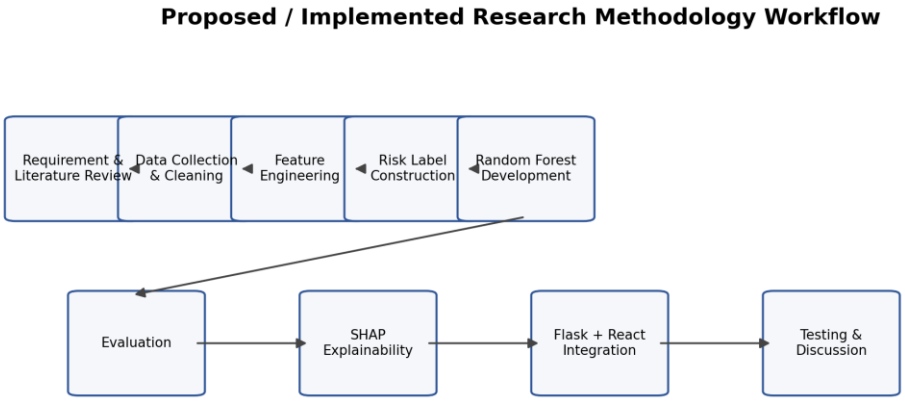


Figure 3.1: Research Methodology Workflow

3.2 Data Participants or Study Materials

No human participant dataset was used to train the risk classifier. The main study materials are the archived stock-market spreadsheet, persisted model artefacts, source code, evaluation images, and live/archived external market data. The model-development spreadsheet contains 33,080 rows covering 1 January 2020 to 31 December 2023 and includes 33 symbols from four country groups. The Sri Lankan subset contains HHL, JKH, and LLUB.

Table 3.1: Dataset Profile

Attribute	Archived dataset value
Raw records	33,080
Date range	2020-01-01 to 2023-12-31
Number of stock symbols	33
USA records	10,060

Attribute	Archived dataset value
UK records	10,080
India records	9,920
Sri Lanka records	3,020
Sri Lankan UI symbols	HHL, JKH, LLUB
Raw global indicators	Gold, Oil, VIX

3.3 Experimental or Study Procedure

Raw price records are ordered by stock and date. Daily return is derived from consecutive closing prices, MA10 and MA50 are calculated as rolling averages, and the model-development Volatility feature is calculated as a ten-period rolling standard deviation of returns. The selected stock symbol is encoded using a persisted LabelEncoder. Gold, Oil, and VIX are aligned with the stock data and used as additional model inputs.

The final input order is fixed because the persisted scikit-learn model expects named columns. The nine model inputs are shown in Table 3.2.

Table 3.2: Model Input Features

Feature	Role in the component
Close	Latest/selected closing or last-trade price supplied to the model.
Volume	Trading volume of the selected stock.
MA10	Ten-period moving average representing short-term price trend.
MA50	Fifty-period moving average representing a longer-term reference.
Volatility	Recent variation; model-development data uses rolling return volatility.
Gold	Latest Gold futures value used as global context.
Oil	Latest crude-oil futures value used as global context.
VIX	Latest VIX value used as market-uncertainty context.
Stock_encoded	Numeric stock-symbol representation produced by the saved LabelEncoder.

The prototype target is constructed from volatility bands. Values below 0.01 are labelled LOW, values from 0.01 up to but not including 0.03 are labelled MEDIUM, and values of 0.03 or above are labelled HIGH. These are operational cut-offs for the prototype rather than universally accepted market-risk boundaries.

Table 3.3: Risk-Class Construction

Prototype rule	Risk label	Interpretation
Volatility < 0.01	LOW	Relatively lower prototype volatility band.
0.01 <= Volatility < 0.03	MEDIUM	Intermediate prototype volatility band.
Volatility >= 0.03	HIGH	Higher prototype volatility band.

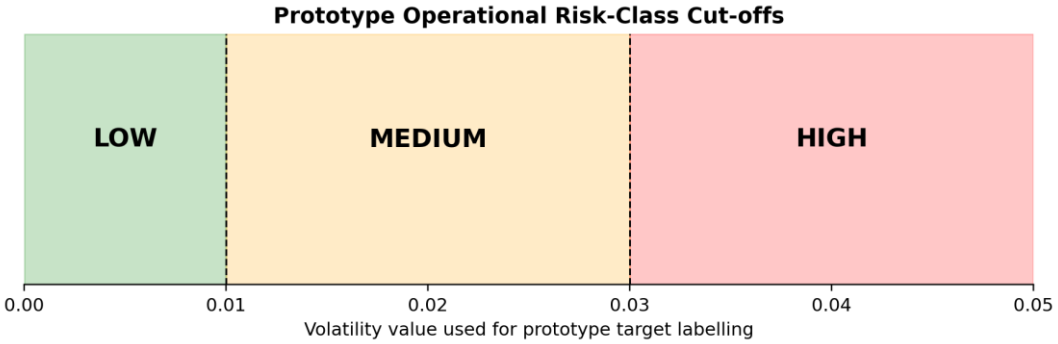


Figure 3.2: Prototype Operational Risk-Class Cut-offs

A key methodological concern follows from this labelling rule: Volatility is used to create the target class and is also included as an input feature. This circular relationship can make the classification task unusually easy and can inflate the apparent evaluation result. The issue is therefore treated as a limitation rather than hidden.

3.4 Measures and Evaluation Metrics

Table 3.4: Evaluation Measures

Measure	Purpose
Accuracy	Overall proportion of correctly classified observations on the reported split.
Precision	Class-specific proportion of predicted cases that

Measure	Purpose
	are correct.
Recall	Class-specific proportion of actual cases that are identified.
F1-score	Harmonic mean of precision and recall.
Confusion matrix	Shows class-by-class correct classifications and misclassifications.
Random Forest feature importance	Provides a model-level view of impurity-based feature use.
SHAP local attribution	Provides class-specific feature contributions for an individual prediction.
Integration checks	Verify data retrieval, inference, explanation, and dashboard delivery.

3.5 Data Analysis

The archived confusion matrix is analysed as evidence of how well the model reproduced the prototype labels on one reported split. Feature importance is analysed to identify whether the model depends strongly on one feature. Historical replay checks for HHL, JKH, and LLUB are used to confirm that the saved model accepts the intended feature structure. SHAP values are interpreted for the predicted class and ranked by absolute magnitude. The analysis deliberately separates current risk-state classification from future-risk forecasting.

3.6 Ethics Privacy Security and Responsible AI

The system is intended for informational decision support and not as personalised financial advice. The dashboard includes an investment caution, and the language-model prompt is designed not to claim model confidence. Credentials, API keys, and database connection strings should not appear in published code or screenshots; production use should move them to environment variables or a secret-management service. The system should also fail safely when external services are unavailable rather than fabricating market values or explanations.

Responsible AI is addressed by separating prediction from explanation. Random Forest produces the class, SHAP provides feature attribution, and Gemini only

rewrites the structured output. This improves traceability and reduces the chance that a language model silently changes the underlying decision.

3.7 Reproducibility and Research Artefacts

The project archive contains the persisted Random Forest classifier, LabelEncoder, dataset files, source code, and evaluation images. The final training notebook or script is not present in the supplied archive, so the exact final train/test split cannot be independently reconstructed from the current package alone. For full reproducibility, the final submission archive should include the training code, dataset version, package versions, split strategy, and random seed together with the persisted artefacts.

4 PRODUCT AND SOLUTION DESIGN

4.1 Requirements

Table 4.1: Requirements

Category	Requirement
Functional	Select a supported CSE stock and retrieve current Close and Volume information.
Functional	Calculate MA10, MA50, and a recent volatility measure from historical records.
Functional	Retrieve Gold, Oil, and VIX automatically.
Functional	Classify the selected stock as LOW, MEDIUM, or HIGH risk.
Functional	Identify influential factors using SHAP.
Functional	Generate a plain-language explanation without allowing the LLM to determine the class.
Functional	Return the integrated result through Flask and display it in React.
Non-functional	Keep prediction and explanation responsibilities separated.
Non-functional	Protect credentials and connection strings.
Non-functional	Use consistent feature definitions between training and inference.
User	Present stock metrics, risk class, global indicators, factors, and explanation in one view.
User	State that the output is informational and not guaranteed financial advice.

4.2 Component Architecture within the Complete Solution

The component is organised as layered services. CSE current data provides stock-level values, MongoDB holds historical records used for moving-average and volatility calculations, and yfinance provides the selected global indicators. The Flask service prepares the nine-feature input and calls the persisted Random Forest model. SHAP processes the model result, Gemini converts the structured explanation into plain language, and the React RiskDashboard presents the integrated output.

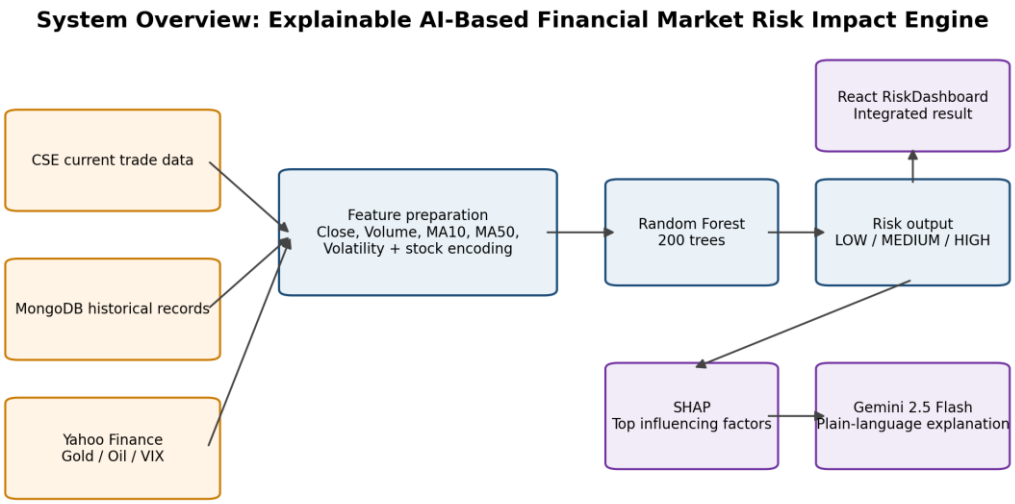


Figure 4.1: Component Architecture within the Complete Solution

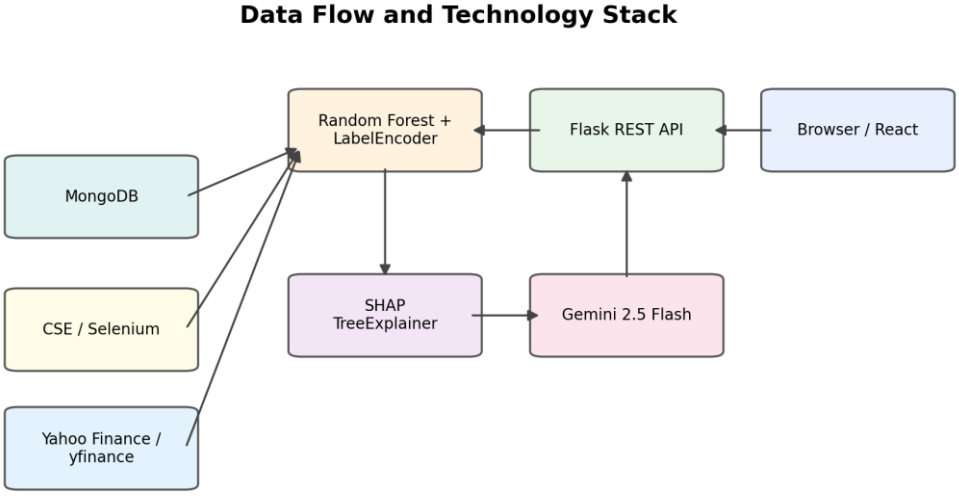


Figure 4.2: Data Flow and Technology Stack

4.3 Design Alternatives and Decisions

Table 4.2: Design Alternatives and Decisions

Design area	Alternative considered	Selected decision and reason
Classifier	Single Decision Tree vs Random Forest	Random Forest selected for ensemble robustness and compatibility with TreeExplainer.
Explanation	Global feature importance only vs local XAI	SHAP selected because the component must explain an individual prediction.
Natural-language layer	LLM decides risk vs LLM explains structured output	LLM restricted to presentation so the ML decision remains traceable.

Design area	Alternative considered	Selected decision and reason
Backend	Node-only vs Python ML service	Flask selected for direct integration with scikit-learn, SHAP, Pandas, and yfinance.
Storage	Flat files only vs database-backed history	MongoDB used by the prototype for historical CSE records and endpoint access.
Frontend	Static report vs interactive dashboard	React/Vite selected for the integrated investor-facing interface.

4.4 Detailed Design

The input contract contains nine features in a fixed order: Close, Volume, MA10, MA50, Volatility, Gold, Oil, VIX, and Stock_encoded. The Flask /predict_auto route receives the stock-specific values, adds the global indicators, applies the encoder, builds the input DataFrame, obtains the model class, maps 0/1/2 to LOW/MEDIUM/HIGH, selects the predicted-class SHAP contributions, ranks the top factors, and sends the structured result to the explanation layer.

For multiclass SHAP output arranged as sample x feature x class, the predicted class must be selected from the class dimension. This detail is important because selecting the wrong axis can produce factors that do not correspond to the displayed risk class. The final implementation should therefore use class-specific indexing consistently.

4.5 Standards and Professional Practice

Professional practice considerations include version control, reproducible environments, input validation, secure secret handling, source acknowledgement, clear figure and table labelling, and explicit communication of uncertainty. The system should not present hard-coded interface values as model confidence. If probabilities are later shown, they should be generated by the model and assessed for calibration before being interpreted as certainty.

5 IMPLEMENTATION INTEGRATION AND TESTING

5.1 Development Environment and Technology Stack

Table 5.1: Technology Stack

Technology	Purpose
Python / Flask	Machine-learning REST service and data-processing endpoints.
scikit-learn / Joblib	Random Forest classifier, LabelEncoder, and model persistence.
Pandas / NumPy	Tabular processing and numerical calculations.
SHAP	Local feature attribution for Random Forest predictions.
yfinance	Retrieval of Gold futures, crude-oil futures, and VIX values.
MongoDB	Historical CSE records used by feature endpoints.
Selenium	Automated CSE trade-summary CSV retrieval in the prototype.
React / Vite	Investor-facing RiskDashboard.
Gemini 2.5 Flash	Plain-language explanation layer.
GitHub	Version control and team integration.

5.2 Component Implementation

The Flask service loads the persisted model and LabelEncoder at start-up. Separate endpoints provide market data and technical features, while /predict and /predict_auto provide inference. /predict_auto is the main integrated path: it supplements the submitted stock-specific values with Gold, Oil, and VIX, runs the model, derives the SHAP factors, requests the plain-language explanation, and returns a JSON response to the frontend.

Table 5.2: Flask API Endpoints

Endpoint	Method	Purpose
/market-data	GET	Returns current Gold, Oil, and VIX values.
/download-cse	GET	Downloads CSE trade-summary data through Selenium in the prototype.

Endpoint	Method	Purpose
/cse-filter	GET	Filters selected companies and stores current records in MongoDB.
/ma10/<symbol>	GET	Calculates the recent ten-period moving average.
/ma50/<symbol>	GET	Calculates the recent fifty-period moving average.
/volatility/<symbol>	GET	Calculates recent variation in the deployed endpoint.
/predict	POST	Predicts risk when the full feature vector is supplied manually.
/predict_auto	POST	Adds live global indicators, predicts risk, computes factors, and generates a plain-language explanation.

The React interface maps company names to CSE/model symbols, requests the technical features, and submits the prepared stock values to /predict_auto. The returned object is stored in React state and rendered as the selected stock, risk class, influential factors, Gold, Oil, VIX, and the AI Market Analyst explanation.

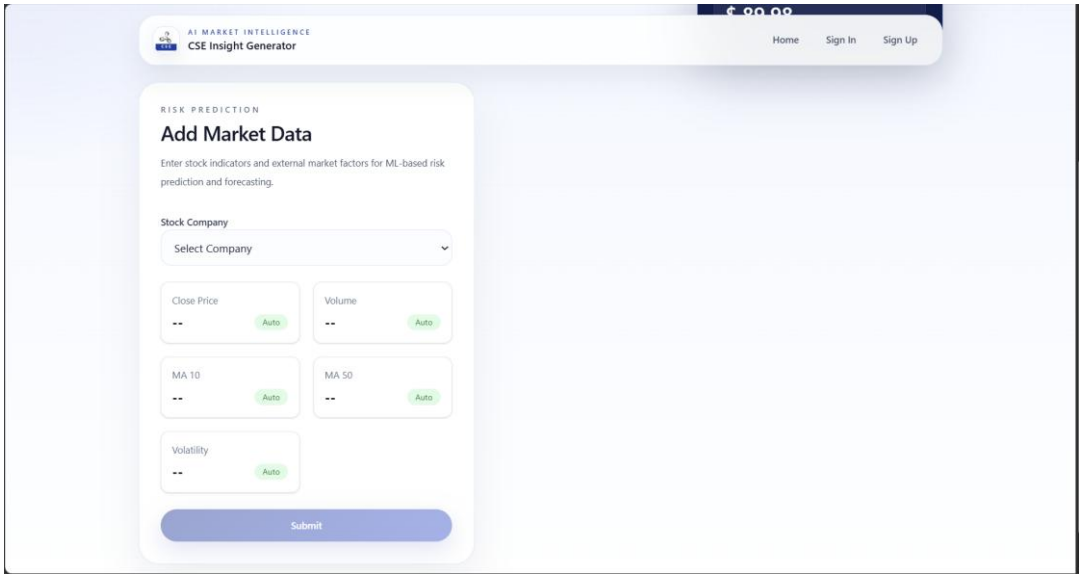


Figure 5.1: Initial Market Data Input Interface

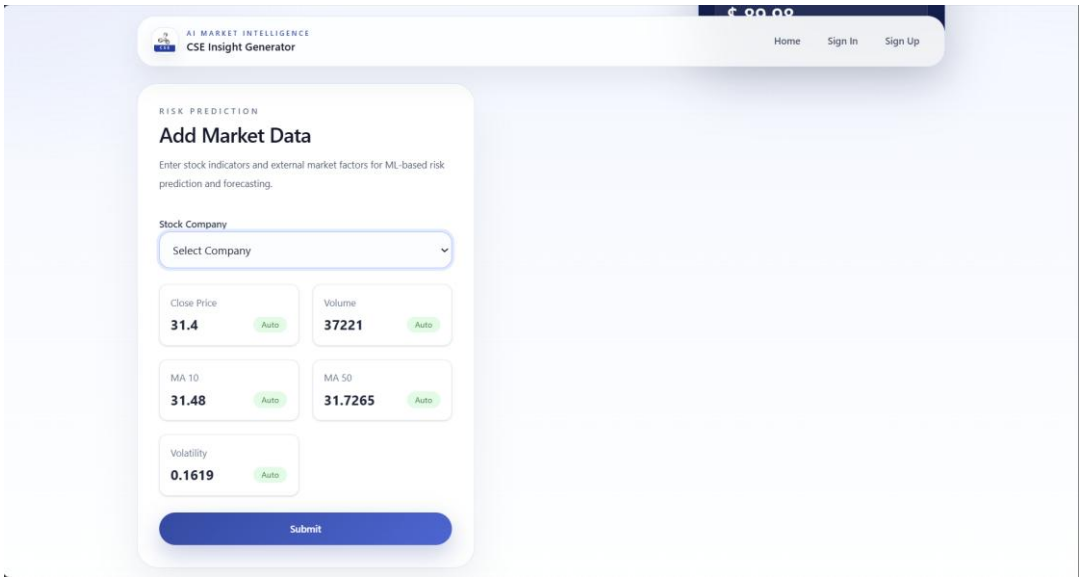


Figure 5.2: Automatic Retrieval of Stock-Specific Indicators

5.3 Integration with the Other Components

Within the wider group application, the risk component exposes an integrated result that can be presented alongside other market-insight functions. The machine-learning service is kept separate from shared platform functions handled by the group backend. This separation reduces coupling: the risk component can be updated or retrained without requiring the entire platform to be redesigned, provided that the API contract remains stable.

5.4 Verification and Software Testing

Table 5.3: Verification and Software Testing

Test / check	Expected result	Evidence / status
Load persisted model and encoder	Artefacts deserialize and expose 9 features / 3 classes.	Verified from archived files.
Select supported CSE stock	Frontend maps HHL, JKH, or LLUB correctly.	Implemented in RiskDashboard.jsx.
Calculate MA10 / MA50	Backend averages recent historical values.	Implemented in Flask endpoints.
Calculate volatility	Backend returns recent variation measure.	Implemented, but live definition differs from model-development convention.
Retrieve Gold / Oil / VIX	Backend requests GC=F, CL=F, and ^VIX.	Implemented; live network verification required.
Predict LOW / MEDIUM / HIGH	Model returns class 0/1/2 and maps to label.	Verified from persisted model and mapping.

Test / check	Expected result	Evidence / status
Return SHAP factors	API returns class-specific top factors.	Concept implemented; class indexing must be correct.
Generate explanation	Gemini converts structured output to plain language.	Implemented; requires live API availability.
Render integrated dashboard	React displays result and caution text.	Demonstrated by current screenshots.

5.5 Deployment and Operation

The present system is a research prototype rather than a production deployment. Operational use would require secure environment-variable management, reliable hosting for the Flask and web layers, controlled database access, logging, monitoring, rate-limit handling, and fallbacks for external services. The yfinance, CSE, Selenium, MongoDB, and Gemini dependencies should be treated as potentially unavailable, and the interface should report a clear error rather than substituting stale or fabricated values.

6 RESULTS AND VALIDATION

6.1 Research Results

The archived evaluation artefact contains a confusion matrix with 445 LOW cases, 4,232 MEDIUM cases, and 1,614 HIGH cases on the diagonal and no off-diagonal errors. This corresponds to 6,291 evaluated observations and an observed accuracy of 100% on that reported split. Under the same matrix, precision, recall, and F1-score are also 1.0 for each class.

Table 6.1: Confusion Matrix Values

Actual \ Predicted	LOW	MEDIUM	HIGH
LOW	445	0	0
MEDIUM	0	4,232	0
HIGH	0	0	1,614

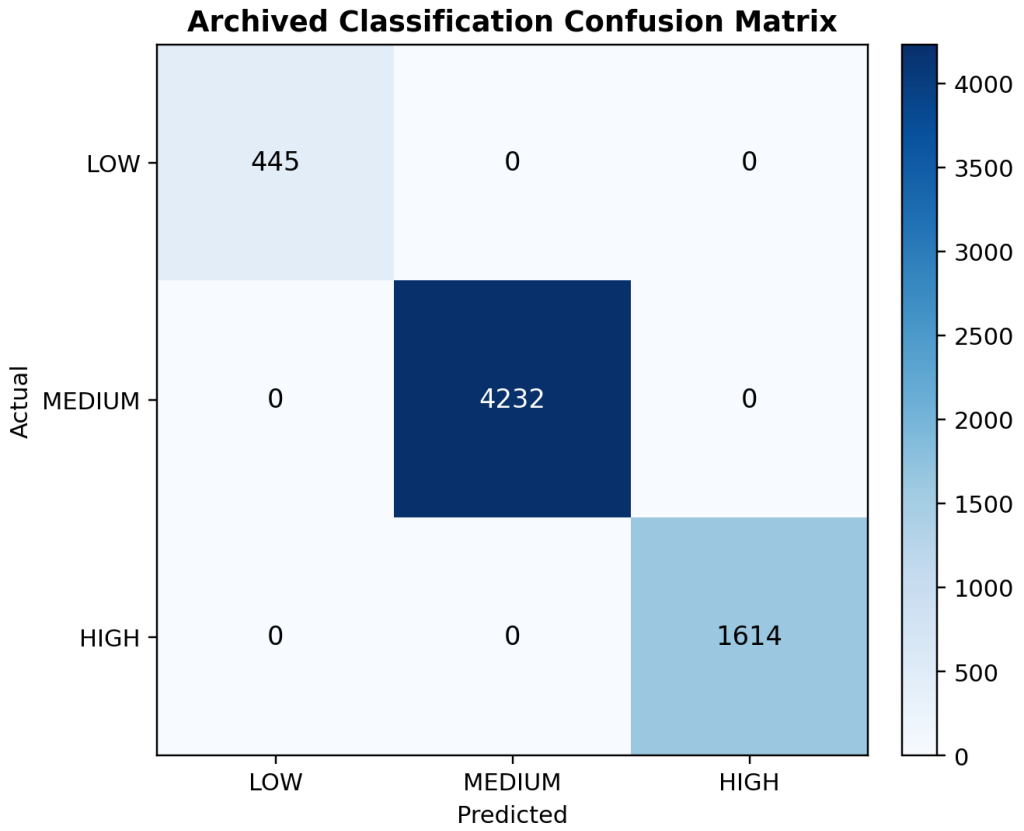


Figure 6.1: Archived Classification Confusion Matrix

The result must not be interpreted as proof of perfect market-risk prediction. Because the target classes are created from volatility bands and Volatility is also provided as an input, the model can reproduce a boundary that is already present in the feature set. The confusion matrix therefore shows that the classifier learned the current prototype rule on the reported split, not that it can forecast future market losses with perfect accuracy.

6.2 Product and Technical Results

The persisted Random Forest model assigns approximately 0.8818 of its impurity-based feature importance to Volatility, followed by Volume at approximately 0.0571 and VIX at approximately 0.0215. The dominance of Volatility is consistent with the target construction and provides additional evidence that the classifier is learning the volatility-defined label boundary.

Table 6.2: Feature Importance

Feature	Importance	Share
Volatility	0.881841	88.18%
Volume	0.057052	5.71%
VIX	0.021491	2.15%
Oil	0.011457	1.15%
Close	0.008397	0.84%
MA10	0.006272	0.63%
Stock_encoded	0.005513	0.55%
MA50	0.004407	0.44%
Gold	0.003570	0.36%

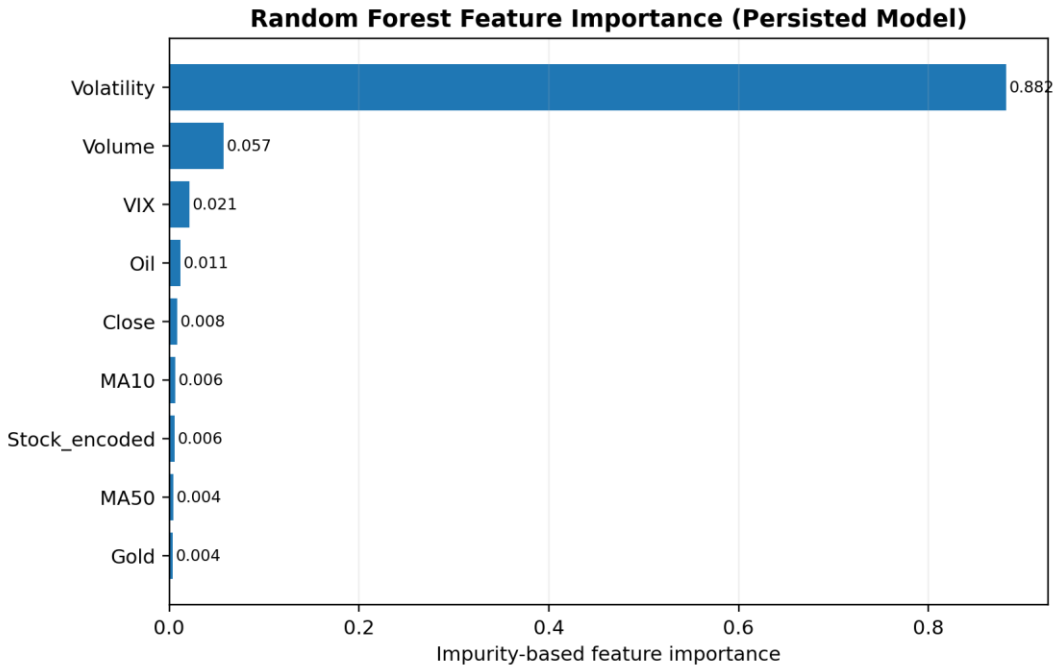


Figure 6.2: Random Forest Feature Importance (Persisted Model)

Historical replay checks were carried out for the latest available HHL, JKH, and LLUB observations in the archived dataset. These checks confirm that the saved model accepts the intended feature structure and returns a class for the three CSE symbols supported by the dashboard. They are reproducibility checks rather than live investment recommendations.

Table 6.3: Historical Replay Examples

Stock	**Historical date**	**Return volatility**	**Predicted risk**	**Top class-specific SHAP factors**
HHL	2023-12-31	0.0146	MEDIUM	Volatility (+0.3190), MA10 (-0.0070), Gold (-0.0048)
JKH	2023-12-29	0.0040	LOW	Volatility (+0.8027), Volume (+0.0240), VIX (+0.0211)
LLUB	2023-12-29	0.0093	LOW	Volatility (+0.7832), Volume (+0.0338), VIX (+0.0169)

6.3 User or Stakeholder Validation

A formal human-subject usability or comprehension study was not included in the supplied research artefacts. Therefore, this thesis does not claim statistically validated user understanding or usability. The current evidence is limited to the implemented dashboard design, the plain-language explanation output, and the explicit separation between prediction and explanation. A future controlled user study with CSE investors or finance students would be needed to measure whether SHAP-supported plain-language explanations actually improve understanding, trust calibration, or decision quality.

6.4 Integration Results

The current system screenshots demonstrate the complete user-facing path from selecting a stock to receiving a risk result and explanation. For HHL, the interface shows a HIGH RISK result together with SHAP-derived market influencers, Gold, Oil, VIX, and the AI Market Analyst explanation. The displayed confidence percentage is an interface placeholder in the current prototype and is not treated as a model-evaluation result.

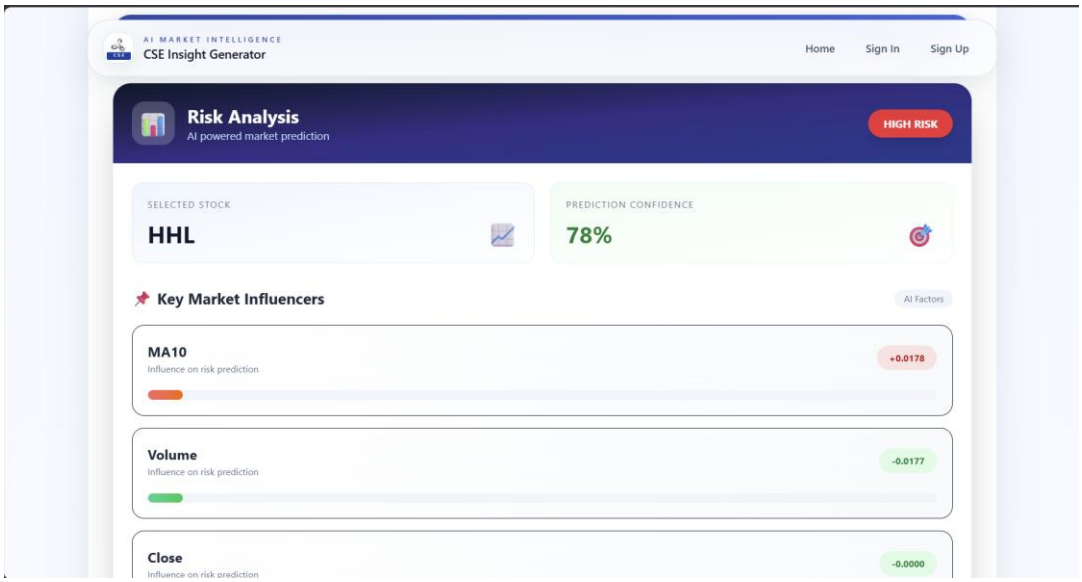


Figure 6.3: Risk Classification Output for HHL

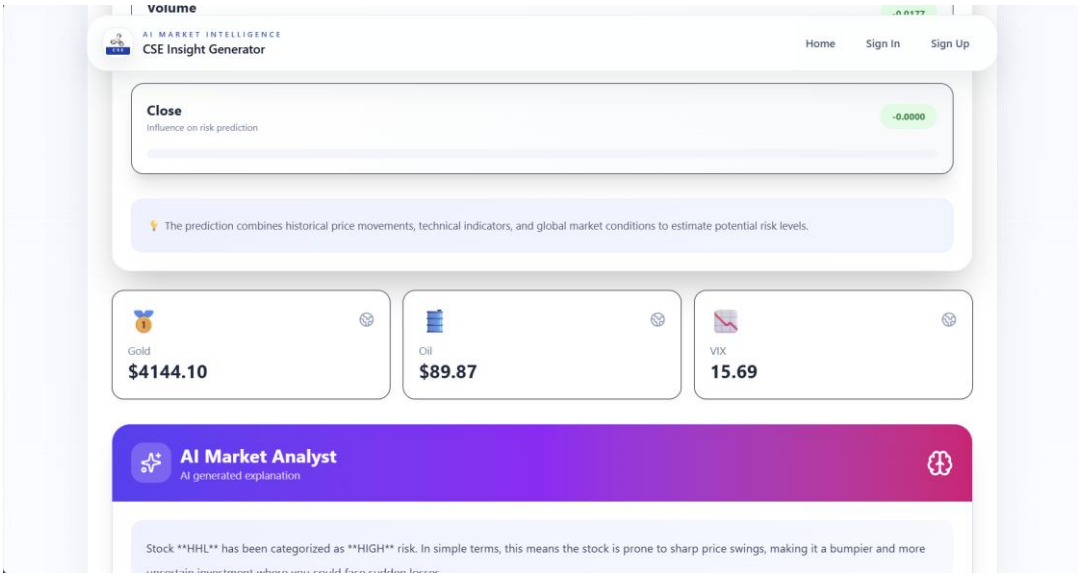


Figure 6.4: Global Financial Indicators and AI Explanation

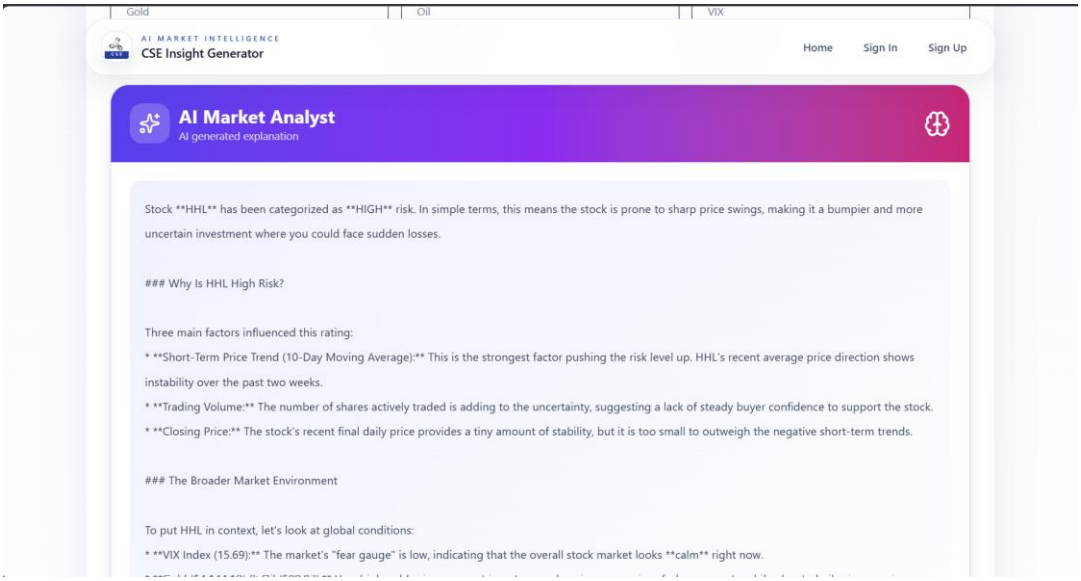


Figure 6.5: AI Market Analyst Explanation of the Risk Result

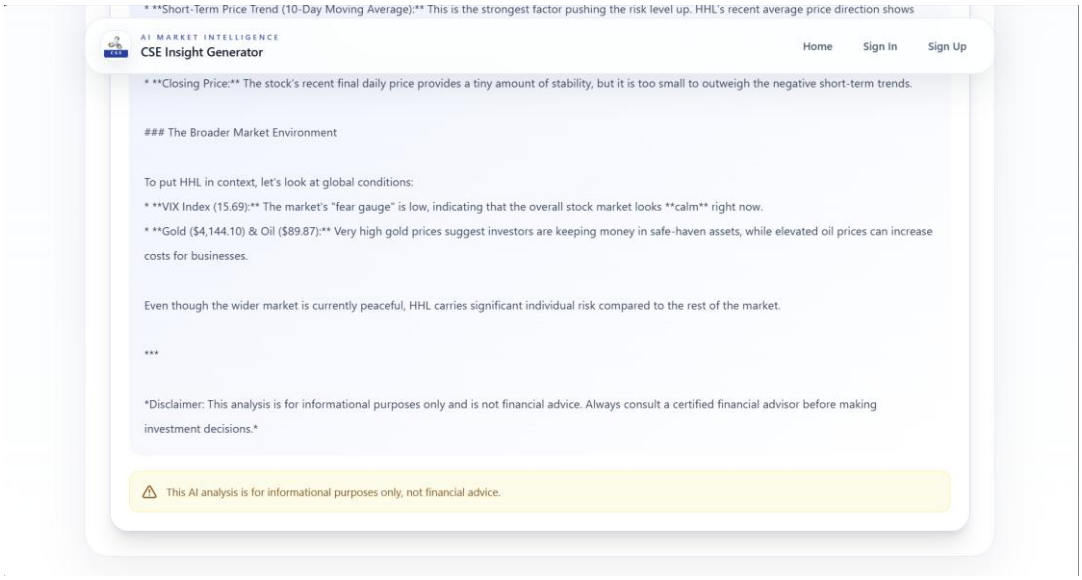


Figure 6.6: Continuation of the AI Explanation and Investment Caution

6.5 Summary of Findings

Table 6.4: Summary of Findings

Finding	Evidence	Interpretation
End-to-end integration is feasible	CSE data, MongoDB, yfinance, Random Forest, SHAP, Gemini, and React are connected.	The component demonstrates a complete explainable-risk workflow.
Random Forest learned the prototype labels extremely well	Archived confusion matrix has zero off-diagonal errors.	Shows learnability of the current rule, not future-risk forecasting.
Volatility dominates the persisted model	Impurity importance is approximately 88.18%.	Reflects the strong link between target construction and input feature.
Local explanations can be produced	Class-specific SHAP extraction identifies top factors.	Provides a useful explanation layer when indexed correctly.
Prediction and explanation can be separated	Random Forest determines class; Gemini receives structured output.	Improves traceability and reduces the chance of LLM decision drift.
Methodological strengthening is required	Target design, multi-market data, preprocessing mismatch, and external dependencies remain.	Future-period targets and chronological validation are required for stronger validity.

7 DISCUSSION

7.1 Interpretation of Findings

The most striking quantitative result is the perfect confusion matrix. In a normal predictive study, such a result would require immediate checks for leakage, duplication, overfitting, or an overly simple target. In this project the main reason is visible in the methodology: risk labels are created from volatility, and Volatility is then provided to the classifier. The model therefore has direct access to a variable that strongly defines the class boundary. The approximately 88% feature-importance share for Volatility supports this interpretation.

For that reason, the current system is more accurately described as a current risk-state classifier built around an operational volatility definition than as a validated future-risk forecasting model. This wording is important because it keeps the research claim aligned with the evidence actually available.

7.2 Comparison with Literature Baselines and Existing Solutions

The work differs from a purely predictive study because it treats interpretation and user-facing delivery as first-class parts of the system. Random Forest provides the classification capability described by Breiman [3], while SHAP provides the local explanation mechanism proposed by Lundberg and Lee [6]. The component also follows the interpretability motivation discussed by Doshi-Velez and Kim [4] and Molnar [5] by presenting reasons rather than only a class label. However, the current prototype does not provide a rigorous algorithmic baseline comparison against Logistic Regression, XGBoost, or another classifier because the archived final training script is not available in the supplied package.

7.3 Research Contribution

The research contribution is an integrated explainable classification workflow for selected CSE stocks. It demonstrates how stock-specific features, selected global indicators, a persisted classifier, local SHAP attribution, and a plain-language explanation layer can be organised so that the risk decision remains traceable. The work also contributes a transparent analysis of its own methodological weakness

rather than treating the 100% split result as evidence of production-level prediction performance.

7.4 Product Contribution

The product contribution is the working risk-dashboard flow. A user selects a supported company, receives automatically populated stock indicators, submits the request, and sees the risk class, influential factors, selected global indicators, and an explanatory narrative in one interface. This creates a clearer path from model output to user interpretation than a standalone notebook or terminal result.

7.5 Limitations and Threats to Validity

The main internal-validity threat is target leakage or circularity caused by using Volatility both to construct the label and as a model input. A second limitation is the preprocessing inconsistency between the model-development volatility definition (rolling standard deviation of returns) and the current live endpoint (standard deviation of recent price values). A third limitation is the use of multi-market development data for a CSE-facing interface. A fourth limitation is the missing final training notebook, which prevents full independent reproduction of the reported split. External dependencies such as Yahoo Finance, CSE retrieval, MongoDB, Selenium, and Gemini also affect operational reliability.

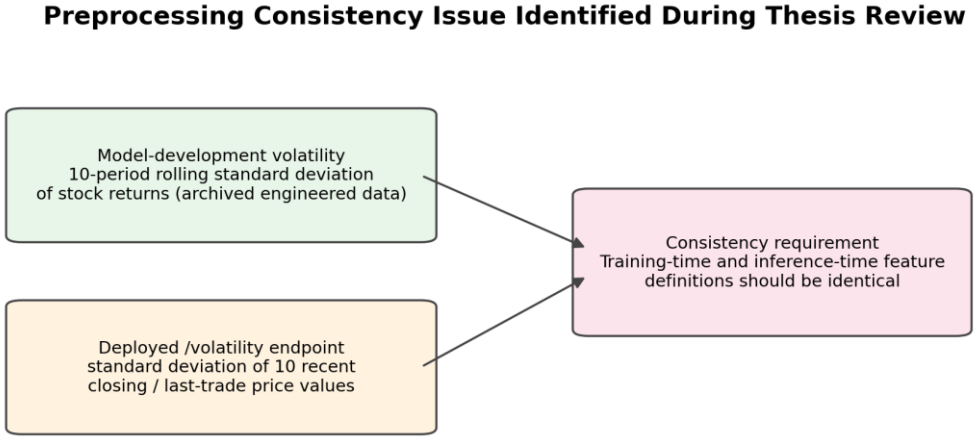


Figure 7.1: Preprocessing Consistency Issue Identified During Thesis Review

The dashboard also contains a fixed confidence percentage. Because that value is not derived from model.predict_proba, it must not be interpreted as scientific

evidence or calibrated certainty. It should be removed or replaced with genuine model-derived probabilities before a final production-oriented release.

7.6 Implications

The main implication is that explainability can be engineered as a separate layer without allowing a language model to determine the financial risk class. This separation is valuable for auditability. Methodologically, the study also shows why high accuracy should be interpreted together with label construction and feature design. A stronger next-stage study would ask whether present-day stock and global indicators can predict a risk measure observed in a future period.

8 COMMERCIALISATION AND SUSTAINABILITY

8.1 Value Proposition and Target Users

The main value proposition is an investor-facing risk-screening component that pairs a simple risk category with explanatory factors and plain-language context. Potential target users include beginner and intermediate CSE investors, financial-education platforms, fintech applications, and brokerage platforms that need an explanatory layer rather than a raw model output.

8.2 Market and Competitive Position

The component is not positioned as a replacement for professional financial analysis. Its differentiating idea is the combination of local CSE-oriented risk presentation, SHAP-based explanation, and a plain-language narrative in one workflow. Commercial positioning would depend on stronger CSE-specific validation, comparison against alternative models, and compliance review before any investment-facing claims are made.

8.3 Business or Sustainability Model

Possible delivery models include a subscription dashboard, institutional licensing, or API access for partner applications. A sustainable service would need to account for data-source costs, model maintenance, monitoring, external-API usage, security, and user support. The architecture is modular enough for the risk engine to be offered as an API, but production use would require service-level monitoring and fallback strategies.

8.4 Scalability Intellectual Property and Compliance

Technical scalability can be improved by separating data acquisition, inference, explanation, and frontend delivery into independently deployable services. Intellectual-property protection would mainly concern the system integration, curated data pipeline, prompt design, interface, and project code rather than ownership of Random Forest or SHAP. Compliance considerations include secure handling of credentials, responsible presentation of financial information, data-source licensing terms, and avoidance of misleading claims of guaranteed returns or certainty.

8.5 Contribution to the Sustainable Development Goals

The component can be linked cautiously to Sustainable Development Goal 8 (Decent Work and Economic Growth) through financial-literacy and decision-support use, and to Goal 9 (Industry, Innovation and Infrastructure) through the development of transparent digital financial infrastructure. These links describe potential alignment rather than measured SDG impact.

Table 8.1: Commercialisation and Sustainability Considerations

Area	**Current position**	**Requirement for stronger commercial readiness**
Value proposition	Explainable risk screening for selected CSE stocks.	Demonstrate measurable user benefit and robust predictive validity.
Scalability	Layered Flask/API + React architecture.	Production hosting, monitoring, caching, and resilient data services.
Security	Research prototype.	Environment variables, secret rotation, authentication, access control, logging.
Compliance	Informational disclaimer included.	Review relevant financial, data, and platform obligations before deployment.
Sustainability	Modular component can be maintained independently.	Budget for external APIs, model review, data updates, and user support.

9 CONCLUSION AND FUTURE DIRECTIONS

9.1 Conclusion

This thesis presented an Explainable AI-Based Financial Market Risk Impact Engine as an individual component of the AI Powered Stock Market Insight Generator for the Colombo Stock Exchange. The system integrates stock-specific and selected global indicators, uses a persisted Random Forest classifier to produce a LOW/MEDIUM/HIGH current risk state, applies SHAP for feature-level explanation, and uses Gemini 2.5 Flash only to rewrite the structured result in simple language. Flask, MongoDB, yfinance, Selenium, and React/Vite are integrated to deliver the workflow through a web interface.

The available evaluation artefact shows that the classifier reproduces the prototype labels perfectly on one reported split, while Volatility accounts for approximately 88.18% of the persisted model's impurity-based feature importance. These results are consistent with the prototype design, but they should not be presented as evidence of perfect future-risk forecasting because the target classes are created from volatility and Volatility is also an input feature.

9.2 Achievement of Objectives

Objective 1: To collect and prepare historical and current stock-market data required for risk analysis. — Addressed in the prototype through the implemented data, model, explanation, API, and dashboard workflow, subject to the limitations documented in Chapters 6 and 7.

Objective 2: To engineer Close Price, Volume, MA10, MA50, and Volatility as stock-specific features. — Addressed in the prototype through the implemented data, model, explanation, API, and dashboard workflow, subject to the limitations documented in Chapters 6 and 7.

Objective 3: To integrate Gold Price, Oil Price, and VIX as selected global market indicators. — Addressed in the prototype through the implemented data, model, explanation, API, and dashboard workflow, subject to the limitations documented in Chapters 6 and 7.

Objective 4: To implement a Random Forest classifier for LOW, MEDIUM, and HIGH risk-state classification. — Addressed in the prototype through the implemented data, model, explanation, API, and dashboard workflow, subject to the limitations documented in Chapters 6 and 7.

Objective 5: To apply SHAP to identify influential features for individual predictions. — Addressed in the prototype through the implemented data, model, explanation, API, and dashboard workflow, subject to the limitations documented in Chapters 6 and 7.

Objective 6: To generate a plain-language explanation from the structured risk and SHAP outputs without allowing the language model to determine the risk class. — Addressed in the prototype through the implemented data, model, explanation, API, and dashboard workflow, subject to the limitations documented in Chapters 6 and 7.

Objective 7: To expose the analytical workflow through a Flask REST API and integrate it with the React/Vite dashboard. — Addressed in the prototype through the implemented data, model, explanation, API, and dashboard workflow, subject to the limitations documented in Chapters 6 and 7.

Objective 8: To evaluate classification behaviour, explainability, implementation quality, integration, and methodological limitations. — Addressed in the prototype through the implemented data, model, explanation, API, and dashboard workflow, subject to the limitations documented in Chapters 6 and 7.

9.3 Individual Research and Product Contributions

The individual research contribution is the transparent integration of classification and explanation. The individual product contribution is the functioning risk-dashboard pathway that connects stock selection, automated technical indicators, Random Forest inference, SHAP factors, selected global indicators, and a plain-language explanation. A further contribution is the explicit documentation of target circularity, preprocessing inconsistency, and interface-placeholder issues that need to be resolved before stronger claims are made.

9.4 Future Directions

- Define the target using a future-period measure such as future volatility, downside loss, or drawdown rather than current volatility.
- Use chronological or rolling time-series validation to reduce leakage between past and future observations.
- Align the exact volatility formula and window between training and live inference.
- Compare Random Forest with appropriate baselines such as Logistic Regression, XGBoost, or other gradient-boosting methods.
- Collect and evaluate a larger CSE-specific historical dataset and compare it with the multi-market development model.
- Correct and verify multiclass SHAP indexing in every prediction route.
- Remove hard-coded confidence values or replace them with model-derived, appropriately calibrated probabilities.
- Conduct a user study to test whether explanations improve investor understanding and trust calibration.
- Add robust fallbacks for external data and generative-AI services and strengthen production security.

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GLOSSARY

Term	Definition
Current risk-state classification	Classification of the current prototype risk condition rather than a forecast of future loss.
Feature importance	A model-level measure of how much a feature contributes to impurity reduction across the Random Forest.
Local explanation	An explanation of the factors associated with one specific model prediction.
Operational threshold	A project-defined cut-off used for prototype labelling; not a universal financial standard.
Target leakage / circularity	A situation where the input contains information strongly tied to how the target was created, producing overly optimistic evaluation.
Volatility	A measure of variation in returns or prices, depending on the exact definition used.

APPENDICES

Appendix A: Archived Dataset Structure

The model-development spreadsheet contains the raw columns below. This table distinguishes the source fields from the nine engineered features used by the saved classifier.

Appendix Table A.1: Archived Dataset Structure

Column	Use / Meaning
Price	Adjusted/derived price field retained in source data.
Date	Observation date.
Close	Closing price used in model feature engineering.
High	Daily high, retained in raw data but not used by the persisted model.
Low	Daily low, retained in raw data but not used by the persisted model.
Open	Daily open, retained in raw data but not used by the persisted model.
Volume	Trading volume used by the model.
Stock	Ticker/symbol.
Country	Market grouping.
Gold	Gold market indicator.
Oil	Crude-oil market indicator.
VIX	VIX market indicator.

Appendix B: Academic Integrity and AI Use Disclosure

Generative AI tools were used during the preparation of this thesis for language refinement, organisation of draft material, and assistance with presentation of diagrams and explanations. They were not treated as primary research evidence. Technical statements were checked against the project code, saved model artefacts, datasets, screenshots, and cited literature available to the author. The author remains responsible for the final wording, citations, code, data, analyses, and conclusions. Any prompts, workflows, or additional disclosure required by the module policy should be retained with the final submission records.

Appendix C: Persisted Model Configuration

Appendix Table C.1: Persisted Model Configuration

Parameter	Value
n_estimators	200
max_depth	10
criterion	gini
max_features	sqrt
bootstrap	True
min_samples_split	2
min_samples_leaf	1
random_state	42
Output mapping	0=LOW, 1=MEDIUM, 2=HIGH

Appendix D: Simplified Backend Feature Contract

The simplified contract below records the feature order and class mapping expected by the persisted model.



```
features = [
    "Close",
    "Volume",
    "MA10",
    "MA50",
    "Volatility",
    "Gold",
    "Oil",
    "VIX",
    "Stock_encoded"
]

risk_map = {0: "LOW", 1: "MEDIUM", 2: "HIGH"}

explainer = shap.TreeExplainer(model)
```



```
        close = float(data["close"])
        volume = float(data["volume"])
        ma10 = float(data["ma10"])
        ma50 = float(data["ma50"])
        volatility = float(data["volatility"])
        gold = float(data["gold"])
        oil = float(data["oil"])
        vix = float(data["vix"])
```



```
        input_df = pd.DataFrame([[
            close,
            volume,
            ma10,
            ma50,
            volatility,
            gold,
            oil,
            vix,
            stock_encoded
        ]], columns=features)

        # ======================================
        # PREDICTION
        # ======================================

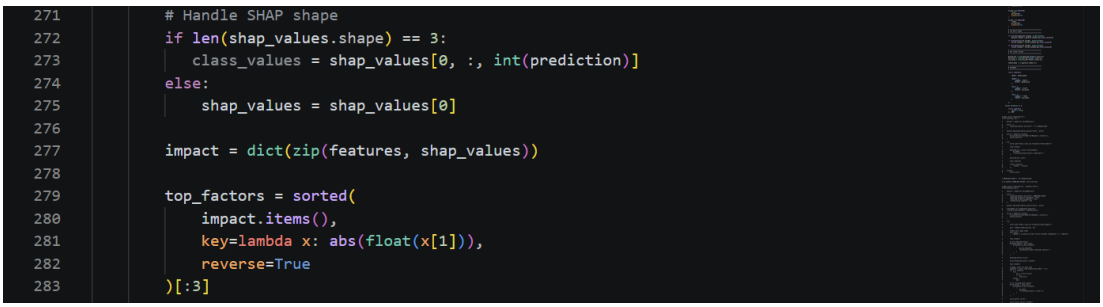
        prediction = mod
        risk = risk_map[prediction]
```

(variable) prediction: Any

The backend constructs the model input using the same nine features and feature order used by the persisted Random Forest classifier. The selected stock symbol is encoded numerically, the feature values are arranged into a Pandas DataFrame, and the classifier returns a numeric class. The numeric prediction is then mapped to LOW, MEDIUM, or HIGH using the defined risk-class mapping.

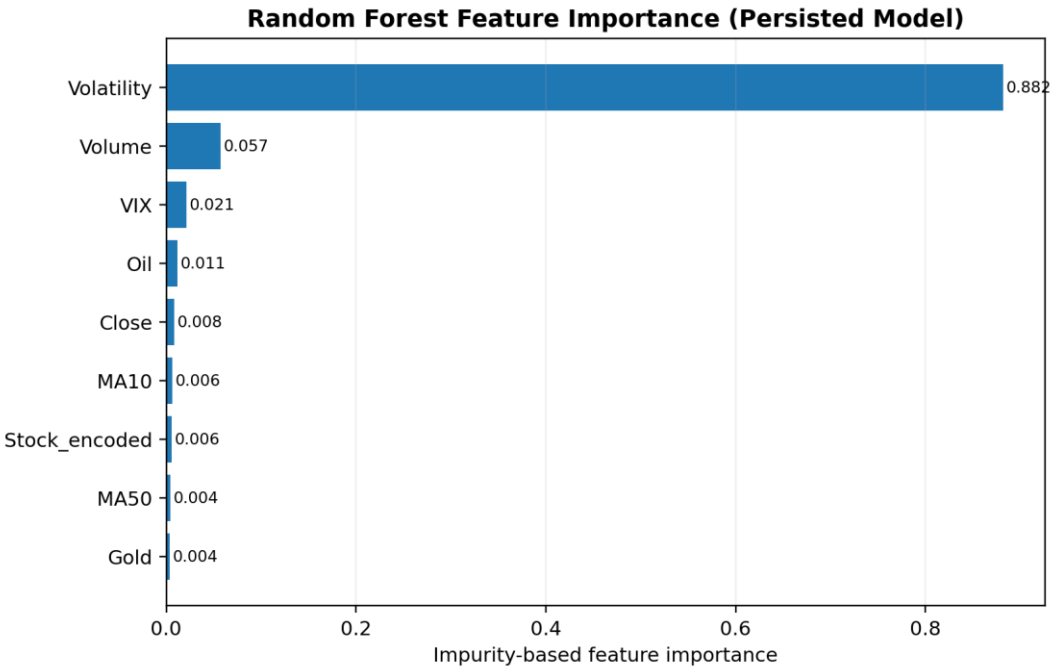
Appendix E: Correct Multiclass SHAP Selection

For a multiclass explanation arranged as sample x feature x class, the predicted class should be selected from the class dimension before the factors are ranked.

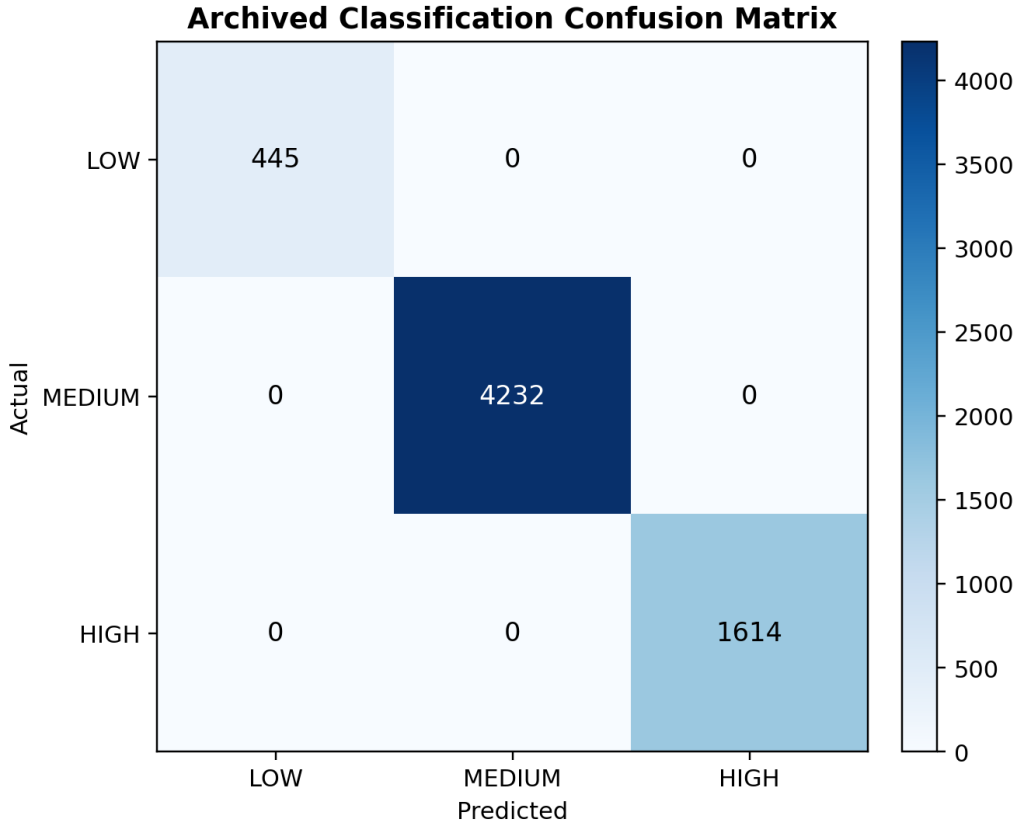


```
271         # Handle SHAP shape
272         if len(shap_values.shape) == 3:
273             class_values = shap_values[0, :, int(prediction)]
274         else:
275             shap_values = shap_values[0]
276
277         impact = dict(zip(features, shap_values))
278
279         top_factors = sorted(
280             impact.items(),
281             key=lambda x: abs(float(x[1])),
282             reverse=True
283         )[:3]
```

Appendix F: Archived Evaluation Artefacts

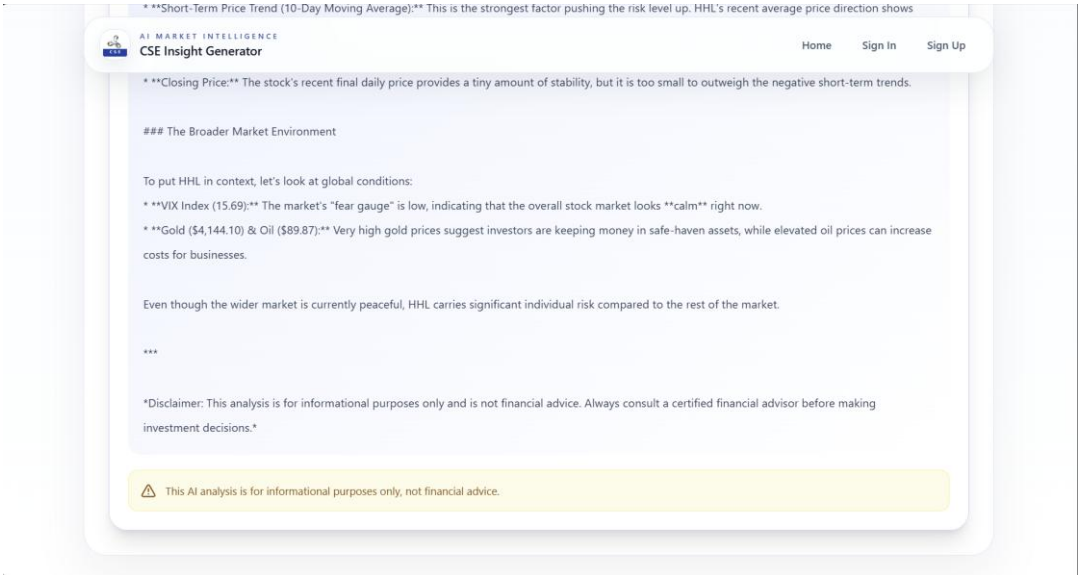


Appendix Figure F.1: Persisted Random Forest Feature Importance



Appendix Figure F.2: Archived Confusion Matrix

Appendix G: Additional Interface Evidence



Appendix Figure G.1: Full AI Explanation Continuation and Disclaimer